

Linux & the Command Line

Servers, cloud and security run on Linux · Equigrow short notes

Moving around

```
pwd                # where am I?
ls -la             # list files, including hidden
cd folder          cd ..          cd ~
mkdir notes        touch a.txt
cp a.txt b.txt     mv a.txt old.txt  rm file    rm -r folder
```

Reading & searching

```
cat file           less file       head -n 5 file     tail -f log.txt
grep "error" app.log # lines containing error
find . -name "*.py" # files by name
wc -l file         # count lines
```

Permissions

ls -l shows e.g. -rwxr-xr--: owner, group, others. r = read (4), w = write (2), x = execute (1).

```
chmod 755 script.sh # owner rwx, others r-x
chmod +x script.sh
sudo command        # run as administrator (careful!)
```

Pipes, processes & networking

```
cat app.log | grep error | wc -l # | sends output to the next command
command > out.txt (overwrite)    command >> out.txt (append)
ps aux      top      kill <pid>
ping google.com  curl https://example.com  ssh user@server
```

A tiny shell script

```
#!/bin/bash
for f in *.txt; do
    echo "Lines in $f: $(wc -l < "$f")"
done
```